INCH-POUND

A-A-59337
16 September 1998
SUPERSEDING
MIL-I-43684B
30 June 1989

COMMERCIAL ITEM DESCRIPTION

INTRENCHING TOOL, HAND, FOLDING, LIGHTWEIGHT

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers a lightweight, foldable, hand intrenching tool.

1.2 Intended use. This intrenching tool is intended for use as a shovel or pick to move dirt, sever roots and clear brush.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U. S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5120

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2. SALIENT CHARACTERISTICS

2.1 Design. The intrenching tool shall have a foldable curved blade and "D" type handle. The tool shall be capable of being used as a shovel when fully extended and as a pick with the shovel blade positioned at a 90° angle. The shovel blade and handle shall be capable of being locked. The edges of the shovel blade shall be beveled. The intrenching tool shall have one serrated cutting blade edge and one axe blade edge. The edges of the shovel blade shall be grounded to a 35° ± 3° angle. When folded, the intrenching tool envelope dimensions shall be 9-1/4"L x 6-1/16"W x 2-1/4"H (± 1/16 inch). The extended length of the intrenching tool shall be 23-1/4 ± 1/16 inches. The intrenching tool shall interface with the intrenching tool carrier conforming to MIL-C-43818B; Carrier, Intrenching Tool, Hand, Folding, Lightweight, Plastic, LC-1, National Stock Number (NSN) 8465-01-001-5474.

2.2 Interface requirements. The folded intrenching tool, when placed in the carrier, shall have a snug fit when examined as specified in 4.3.1. The intrenching tool carrier shall be capable of being closed and fastened with the intrenching tool placed in the carrier when tested as specified in 4.3.1

2.3 Physical requirements.

2.3.1 Thrust washer and nut bonding. The thrust washer and nut subassembly shall be completely bonded and shall be tested as specified in 4.3.2.1.

2.3.2 Hardness. The shovel blade shall have a Rockwell "C" hardness within the range of 40-46. Examination of the hardened blade shall be as specified in 4.3.2.2.

2.3.3 Finish. The intrenching tool shall have a chip and scratch-resistant finish.

2.3.3.1 Steel components. Prior to assembly, steel components shall be either phosphate coated or pretreated for painting and shall be coated with enamel having a minimum thickness of 0.0015 inch (see 6.5) . Examination of the enamel shall be as specified in 4.3.2.3.

2.3.3.2 Aluminum components. Prior to assembly, the aluminum components shall be anodized using standard commercial practice (see 6.5). The anodized components shall be examined as specified in 4.3.2.3.

2.3.4 Corrosion resistance. The intrenching tool shall be made from corrosion-resistant material. The intrenching tool shall be tested for corrosion resistance as specified in 4.3.3.1.

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2.3.5 Weight. The weight of the intrenching tool shall not exceed 2.5 pounds when examined as specified in 4.3.3.2.

2.3.6 Color. The color of the intrenching tool shall conform to color number 37038 of Federal Standard 595.

2.3.7 Lubrication. All threaded areas and working joints shall be treated with a light coating of Dow Chemical Molycoat number 557, or equivalent.

2.4 Marking. Each intrenching tool shall be permanently and legibly marked with the name of the manufacturer and the year of manufacture. The letters U.S. shall be permanently and legibly marked on the intrenching tool handle.

2.5 Workmanship. The surfaces of the intrenching tool shall be free of burrs, slivers, sharp corners, and rough or sharp edges (except for the cutting edges of the shovel blade). Feathered edges shall be removed from the shovel blade. Coatings shall be continuous with no missed coverage, smooth, dry film and shall be free of any embedded foreign matter. Rivets shall completely fill the rivet holes and rivet heads shall be full, neatly made, concentrical with the rivet hole and shall be in full contact with the riveted member.

2.6 Performance requirements.

2.6.1 Blade folding. The shovel blade shall be capable of being folded on its hinges to the pick and shovel positions with a weight load of not less than 2 pounds and not more than 7 pounds when tested as specified in 4.3.3.3.

2.6.2 Strength. The intrenching tool shall not develop a permanent deflection of more than 5/16 inch when tested as specified in 4.3.3.4 (Procedure A). Intrenching tool components shall not fracture or break and the intrenching tool shall not be rendered inoperable when strength tested in accordance with 4.3.3.4 (Procedure B).

2.7 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of ASTM E 380, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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3. REGULATORY REQUIREMENTS

3.1 Contractor-recovered material The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, shall conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market or the same product that has successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such compliance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Market acceptability (MA). The following market acceptability criteria is necessary to document the quality of the product to be provided under this CID:

The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years.

4.3 Tests

4.3.1 Intrenching tool/carrier interface. The intrenching tool and carrier shall be examined and tested for interface by placing a folded intrenching tool inside the carrier. The carrier shall then be folded and fastened. Inability to close and fasten the carrier shall constitute an interface failure. Evidence of a loose fit between the tool and carrier or evidence of a bulge in any section of the carrier shall constitute an interface failure.

4.3.2 Component tests

4.3.2.1 Thrust washer and nut bond test. Thrust washer and nut subassemblies shall be tested by affixing the nut in a vise or fixture and applying 500 pounds of force against the inner face of the washer. Any delamination or separation of the washer from the nut shall constitute a test failure.

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4.3.2.2 Hardness. The intrenching tool blade shall be tested for hardness in accordance with ASTM E18. Three readings will be taken on the blade within a 4-inch triangle. Any value not within the hardness range specified in 2.3.2 shall constitute a test failure.

4.3.2.3 Finish. Conduct a visual examination and verify that all steel components have been completely coated. Using standard commercial practice, verify that enameled components have a minimum coating thickness of 0.0015 inch. Visually verify that aluminum components have been completely anodized.

4.3.3 End item tests.

4.3.3.1 Corrosion-resistance test. The intrenching tool shall be tested for corrosion resistance as specified in ASTM B117. The time of exposure shall be 96 hours. Evidence of corrosion on any component of the intrenching tool shall constitute a test failure.

4.3.3.2 Weight. Verify the weight of the intrenching tool by weighing on a calibrated scale. Failure to meet the requirements of 2.3.5 shall constitute a test failure.

4.3.3.3 Blade folding test. With the blade locking mechanism disengaged and the handle of the intrenching tool held firmly, a load shall be applied perpendicular to the blade 1/2 inch from the tip of the blade. Failure of the blade to fold within the weight range specified in 2.6.1 shall constitute a test failure.

4.3.3.4 Strength test. The intrenching tool shall be strength tested as shown on Figure 1. Failure to meet the requirements specified in 2.6.2 after completion of test Procedure A shall constitute failure of the test. At the completion of test Procedure B, the intrenching tool shall be inspected for evidence of component fracture and loss of functionality. Failure to meet the requirements of 2.6.2 shall constitute a test failure.

5. PACKAGING

5.1 Packaging. Preservation, packing and marking requirements shall be as specified in the contract or purchase order (see 6.1).

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6. NOTES

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of this document.

b. Packaging requirements (see 5.1).

6.2 Sources for non-government association documents:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B117 - Standard Practice for Operating Salt Spray (FOG) Apparatus

E18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

E 380 - Use of the International System of Units (SI) (The Modernized Metric System)

(Application for copies of ASTM standards should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959)

6.3 Source of supply. A manufacturer whose product is known to meet the requirements of this document is listed below; however, competition is not limited to this company.

Ames Company
P. O. Box 1774
Parkersburg, WV 26101

6.4 National Stock Number (NSN). The following NSN is assigned to the intrenching tool:

Intrenching Tool, Hand, Folding, Lightweight - 5120-00-878-5932

6.4.1 Intrenching tool fabrication. The intrenching tool has been fabricated as specified using the following military documents:

Specification - MIL-I-43684B; Intrenching Tool, Hand, Folding, Lightweight

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Drawings -

Drawing No. 2-2-284; Intrenching Tool, Hand, Folding, Lightweight, Complete Assembly

Drawing No. 2-2-285; Intrenching Tool, Hand, Folding, Lightweight - Details A

Drawing No. 2-2-286; Intrenching Tool, Hand, Folding, Lightweight - Details B

Drawing No. 2-2-288; Intrenching Tool, Hand, Folding, Lightweight - Strength and Function Test

6.5 Finishes. Prior to assembly, steel components can be phosphate-coated in accordance with Specification DoD-P-16232, "Phosphate Coatings, Heavy Manganese or Zinc Base (For Ferrous Metals)", followed by a treatment with corrosion-resistant oil. Enamel coating can be accomplished by pretreating the steel components prior to assembly in accordance with TT-C-490" Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings", followed by coating with enamel conforming to TT-E-529, "Enamel Alkyd, Semigloss, Low VOC Content". Prior to assembly, aluminum components can be anodized in accordance with Specification MIL-A-8625, Type II, Class 2, "Anodize Coatings for Aluminum and Aluminum Alloys".

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - SH
Air Force - 99

Review Activities:
Army - MD
Air Force - 84, 85
Navy - AS, CG
DSC - CC

Civil Agency Coordinating Activity:

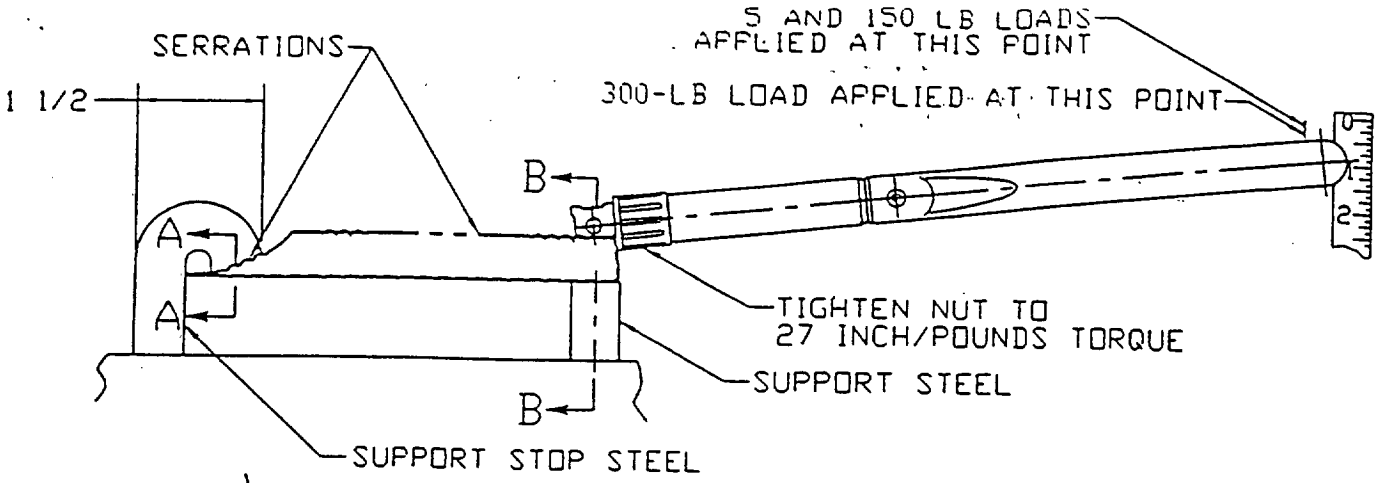
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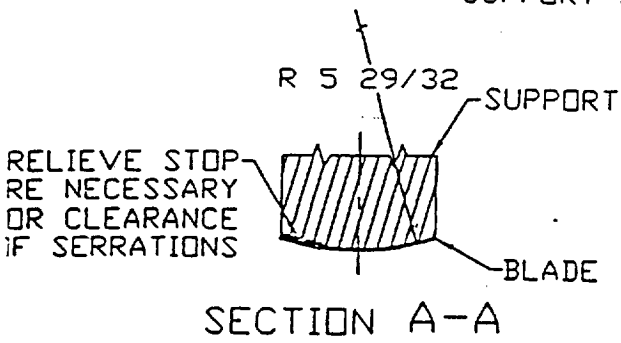
Preparing Activity:

Army - GL

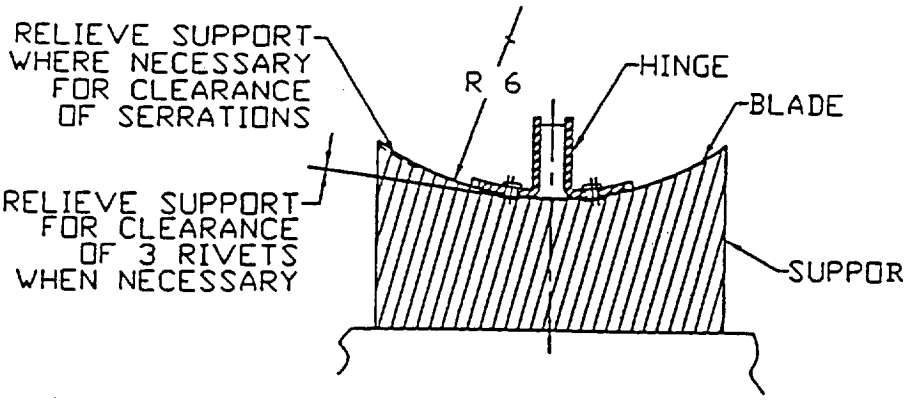
Project No. 5120-0100

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SECTION A-A



SECTION B-B

TEST PROCEDURE:

A. 1. EXTEND TOOL TO TEST POSITION.
2. TIGHTEN HANDLE NUT TO REQUIRED TORQUE.
3. SET TOOL IN JIG.
4. APPLY 5 POUND LOAD.
5. READ AND RECORD SETTING.
6. REMOVE 5 POUND LOAD AND APPLY 150 POUND LOAD.
7. REMOVE 150 POUND LOAD.
8. REMOVE TOOL FROM JIG AND LOOSEN NUT.
9. MOVE BLADE OUT OF AND THEN BACK INTO TEST POSITION.
10. RETORQUE NUT AND SET TOOL BACK IN JIG.
11. APPLY 5 POUND LOAD.
12. READ AND RECORD SETTING.
13. DIFFERENCE BETWEEN FIRST AND SECOND READING IS PERMANENT DEFLECTION.

B. 1. APPLY 300 POUND LOAD.
2. REMOVE TOOL FROM JIG.
3. INSPECT FOR FRACTURE OF ANY COMPONENT.

END ITEM TESTS
FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-59337	2. DOCUMENT DATE (YYMMDD) 98-09-16

3. DOCUMENT TITLE : Commercial Item Description - Intrenching Tool, Hand, Folding, Lightweight

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)
a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial: 508-233-5175 (2) AUTOVON: 256-5175	
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center Attn: SSCNC-WEF Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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